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JOINT FOR CONNECTION TO VARIABLE GUIDE VANE OF GAS TURBINE ENGINE

Abstract

A variable pitch stator vane stage of a gas turbine engine includes a plurality of stator vanes extending across a flowpath of the gas turbine engine, and an actuation assembly operably connected to the plurality of start vanes. The actuation assembly includes an actuation ring extending circumferentially around an engine central longitudinal axis of the gas turbine engine, and a plurality of vane links. Each vane link operably connects a corresponding stator vane of the plurality of stator vanes to the actuation ring. A vane link of the plurality of vane links includes a link body, and a first link end connected the link body. The first link end includes a first spherical connection to the actuation ring. Circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes.

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Background/Summary

BACKGROUND

[0001] Exemplary embodiments of the present disclosure pertain to the art of gas turbine engines. [0002] Gas turbine engines are configured to operate across a wide range of operating conditions, such as speeds, loads, pressures, etc., also referred to as an “operating envelope”. To adapt to these changing operating conditions, many gas turbine engines include variable guide vanes, which are stators that can change their incident angle utilizing an actuator and linkage arrangement to improve performance and operability in gas turbine engines. Similarly, gas turbine engines often include bleed off valves (BOV's) that use a linkage to open and close to bleed airflow from a core flowpath of the engine. The mechanisms which generate movement are often problematic and degrade/wear/fret over time. For example, some configurations use sheet metal arms which must twist during operation, generating high loads and wearing the pin bushing in the drive ring. Other configuration use composite bushings which also wear against their counterpart. One of the main contributors to wear and fretting can be slop/looseness in the system components when coupled with vibration. Wear and fretting in mating components can result in inaccuracy in the positioning of the variable guide vanes and BOV's.

BRIEF DESCRIPTION

[0003] In one exemplary embodiment, a variable pitch stator vane stage of a gas turbine engine, includes a plurality of stator vanes extending across a flowpath of the gas turbine engine, and an actuation assembly operably connected to the plurality of start vanes. The actuation assembly includes an actuation ring extending circumferentially around an engine central longitudinal axis of the gas turbine engine, and a plurality of vane links. Each vane link operably connects a corresponding stator vane of the plurality of stator vanes to the actuation ring. A vane link of the plurality of vane links includes a link body, and a first link end connected the link body. The first link end includes a first spherical connection to the actuation ring. Circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes.

[0004] Additionally or alternatively, in this or other embodiments the first link end includes a first eye, and a first spherical bearing element positioned in the first eye. The vane link is connected to the actuation ring through the first spherical bearing member.

[0005] Additionally or alternatively, in this or other embodiments the first link end a first link fastener extends through the actuation ring and the first spherical bearing member thereby connecting the vane link to the actuation ring.

[0006] Additionally or alternatively, in this or other embodiments a length of the vane link is adjustable.

[0007] Additionally or alternatively, in this or other embodiments the length of the vane link is adjustable via a threaded connection of the link body to the first link end.

[0008] Additionally or alternatively, in this or other embodiments a vane arm is operably connected to a vane of the plurality of vanes and is connected to a second link end of the vane link. The second link end includes a spherical connection to the vane arm.

[0009] Additionally or alternatively, in this or other embodiments the second link end includes a second eye and a second spherical bearing member positioned in the second eye. The vane link is connected to the vane arm through the second spherical bearing member.

[0010] Additionally or alternatively, in this or other embodiments a second link fastener extends through the vane arm and the second spherical bearing member thereby connecting the vane link to the vane arm.

[0011] In another exemplary embodiment, a compressor section of a gas turbine engine includes

one or more rotor stages configured to rotate about an engine central longitudinal axis, and one or more stator stages, the one or more stator stages including a variable pitch stator vane stage. The variable pitch stator vane stage includes a plurality of stator vanes extending across a flowpath of the gas turbine engine, and an actuation assembly operably connected to the plurality of stator vanes. The actuation assembly includes an actuation ring extending circumferentially around the engine central longitudinal axis of the gas turbine engine, and a plurality of vane links, each vane link operably connecting a corresponding stator vane of the plurality of stator vanes to the actuation ring. A vane link of the plurality of vane links includes a link body and a first link end connected to the link body. The first link end includes a first spherical connection to the actuation ring. Circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes.

[0012] Additionally or alternatively, in this or other embodiments the first link end includes a first eye and a first spherical bearing member positioned in the first eye. The vane link is connected to the actuation ring through the first spherical bearing member.

[0013] Additionally or alternatively, in this or other embodiments a first link fastener extends through the actuation ring and the first spherical bearing member thereby connects the vane link to the actuation ring.

[0014] Additionally or alternatively, in this or other embodiments a length of the vane link is adjustable.

[0015] Additionally or alternatively, in this or other embodiments the length of the vane link is adjustable via a threaded connection of the link body to the first link end.

[0016] Additionally or alternatively, in this or other embodiments a vane arm is operably connected to a vane of the plurality of vanes and is connected to a second link end of the vane link. The second link end includes a spherical connection to the vane arm.

[0017] Additionally or alternatively, in this or other embodiments the second link end includes a second eye and a second spherical bearing member located in the second eye. The vane link is connected to the vane arm through the second spherical bearing member.

[0018] Additionally or alternatively, in this or other embodiments a second link fastener extends through the vane arm and the second spherical bearing member thereby connects the vane link to the vane arm.

[0019] In yet another exemplary embodiment, a gas turbine engine includes a combustor section, a turbine section driven by products of the combustion section, a compressor section driven by rotation of the turbine section, and a variable pitch stator vane stage positioned at at least one of the turbine section and the compressor section. The variable pitch stator vane stage includes a plurality of stator vanes extending across a flowpath of the gas turbine engine and an actuation assembly operably connected to the plurality of stator vanes. The actuation assembly includes an actuation ring extending circumferentially around an engine central longitudinal axis of the gas turbine engine and a plurality of vane links, each vane link operably connecting a corresponding stator vane of the plurality of stator vanes to the actuation ring. A vane link of the plurality of vane links includes a link body and a first link end connected to the link body. The first link end includes a first spherical connection to the actuation ring. Circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes.

[0020] Additionally or alternatively, in this or other embodiments the first link end includes a first eye and a first spherical bearing member positioned in the first eye. The vane link is connected to the actuation ring through the first spherical bearing member.

[0021] Additionally or alternatively, in this or other embodiments a first link fastener extends through the actuation ring and the first spherical bearing member thereby connecting the vane link to the actuation ring.

[0022] Additionally or alternatively, in this or other embodiments a length of the vane link is adjustable via a threaded connection of the link body to the first link end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0024] FIG. 1 is a partial cross-sectional view of an embodiment of a gas turbine engine;

[0025] FIG. 2 is a partial cross-sectional view of an embodiment of a variable pitch stator vane;

[0026] FIG. 3 is a partial perspective view of an embodiment of a variable stator vane actuation assembly;

[0027] FIG. 4 is a perspective view of an exemplary embodiment of a vane link;

[0028] FIG. 5 is a partially disassembled view of an exemplary embodiment of a vane link; and

[0029] FIG. 6 is a partially disassembled view of an exemplary embodiment of a vane link.

DETAILED DESCRIPTION

[0030] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0031] FIG. 1 schematically illustrates a gas turbine engine **10**. The gas turbine engine **10** is disclosed herein as a two-spool turbofan that generally incorporates a fan section **12**, a compressor section **14**, a combustor section **16** and a turbine section **18**. Alternative engines might include other systems or features. The fan section **12** drives air along a bypass flow path B in a bypass duct, while the compressor section **14** drives air along a core flow path C for compression and communication into the combustor section **16** then expansion through the turbine section **18**. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines including three-spool architectures.

[0032] The exemplary engine **10** generally includes a low speed spool **20** and a high speed spool **22** mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **26**.

[0033] The low speed spool **20** generally includes an inner shaft **30** that interconnects a fan **32**, and a low pressure turbine **36**. The high speed spool **22** includes an outer shaft **40** that interconnects an impeller **42** and high pressure turbine **44**. A combustor **90** is arranged in exemplary gas turbine **10** between the impeller **42** and the high pressure turbine **44**. The inner shaft **30** and the outer shaft **40** are concentric and rotate about the engine central longitudinal axis A which is collinear with their longitudinal axes.

[0034] The core airflow is compressed by the impeller **42**, mixed and burned with fuel in the combustor **90**, then expanded over the high pressure turbine **44** and low pressure turbine **36**. The turbines **36**, **44** rotationally drive the respective low speed spool **20** and high speed spool **22** in response to the expansion. It will be appreciated that each of the positions of the fan section **12**, compressor section **14**, combustor section **16**, and turbine section **18**, may be varied. While the structure described herein is a two-spool gas turbine engine **10**, one skilled in the art will readily appreciate that the present disclosure may be similarly applied to a single spool or three or more spool gas turbine engine **10**. The core flow path C is separated from the bypass flowpath B by a core casing **46** that encloses the compressor section **14**, the combustor section **16** and the turbine section **18**. The compressor section **14** includes two or more compressor stages, with each compressor stage including at least a compressor rotor **48**. In some embodiments, one or more of the compressor rotors **48** is an impeller.

[0035] One or more of the fan section **12**, the compressor section **14** and the turbine section **18** includes a stator stage **50** including a plurality of variable guide vanes **52**. For clarity of explanation, the present description will describe the stator stage **50** as being included at the compressor section **14**, but one skilled in the art will readily appreciate that the description may be readily adapted to a stator stage **50** positioned at the fan section **12** and the turbine section **18**.

[0036] As stated, the stator stage **50** includes the plurality of variable guide vanes **52**, which are each configured to rotate about their respective vane axis **54**. Referring now to FIG. 2, an embodiment of a vane **52** is illustrated. The vane **52** includes a vane body **56** extending across the core flowpath **C**, and further includes an inner vane stem **58** extending radially inwardly from the vane body **56** and an outer vane stem **60** extending radially outwardly from the vane body **56**. The inner vane stem **58** is positioned at an inner frame **62** and may be supported by an inner bearing **64**. The outer vane stem **60** extends through a vane opening **66** in the core casing **46** and may be supported in the vane opening **66** by an outer bearing **68**.

[0037] Referring now to FIG. 3 with continuing reference to FIG. 2, the vane **52** is connected to an actuation assembly **70** which drives rotation of each of the vanes **52** about the respective vane axes **54** and also controls the rotational position of the vanes **52** relative to their respective vane axes **54**. The outer vane stem **60** protrudes from the core casing **46** and is operably connected to the actuation assembly **70** which is also positioned radially outside of the core casing **46**.

[0038] Each outer vane stem **60** is connected to an actuation ring **72** of the actuation assembly **70** via a vane arm **74** secured to the outer vane stem **60**. In some embodiments, the vane arm **74** is secured to the outer vane stem **60** by a fastener, such as a vane arm bolt **76**. The vane arm **74** is connected to the actuation ring **72** via a vane link **78**. The vane link **78** is secured to each of the actuation ring **72** and the vane arm **74** via link fasteners **80** at each link end **82** of the vane link **78**. The actuation ring **72** is configured to be driven in a circumferential around the core casing **46**. This circumferential movement of the actuation ring **72** urges rotation of each of the vanes **52** about their respective vane axes **54**. The actuation ring **72** is operably connected to an actuator **100**, such as a hydraulic, mechanical, electromechanical or electrical actuator **100** to urge the circumferential movement of the actuation ring **72** relative to the engine central longitudinal axis **A**.

[0039] As illustrated in FIG. 4, and with continuing reference to FIG. 3, the vane link **78** includes a link body **84** and a ball joint or spherical bearing configuration at each link end **82**. The embodiment of FIG. 4 in particular illustrates a spherical bearing configuration, with each link end **82** having an eye **86** with a spherical bearing member **88** located therein. The link fasteners **80** extend at least partially through a bearing opening **98** in the spherical bearing member **88** to connect the vane link **78** to the vane arm **74** and to the actuation ring **72**. The bearing member **88** is movable inside the eye **86**, such that there is a reduced incidence of the link fasteners binding when installed to the bearing member **88**, thus reducing stresses in the actuation assembly **70** and improving operability and accuracy of the positioning of the plurality of vanes **52**. In other embodiments, the link ends **82** may be similarly configured as ball joints.

[0040] Additionally, in some embodiments a link length **92** of the vane link **78** is adjustable to allow for positional adjustment of individual vanes **52** during setup and/or maintenance of the stator stage **50**. The link body **84** may include a body thread **94**, as shown in FIG. 5, with one or more of the link ends **82** having a complimentary end thread **96** via which the link end **82** is installed to the link body **84**. The link length **92** is therefore adjustable by rotating the link end **82** relative to the link body **84**. Nuts **106** lock the vane link **78** to the adjusted link length **92** so that the vane link **78** will not unintentionally shorten or lengthen during operation. While in the embodiment of FIG. 5 the vane link **78** is adjustable at both of the link ends **82**, in other embodiments the link length **92** is adjustable at only one of the link ends **82**.

[0041] While the embodiments illustrated herein include the spherical bearing member **88** in the vane link **78**, in other embodiments the spherical bearing member **88** may be included in the actuation ring **72** and/or in the vane arm **74**.

[0042] In other embodiments, as illustrated in FIG. 6, the vane link 78 includes a ball joint configuration at at least one of the link ends 82. In one embodiment, ball member 102 is secured to via a thread 104, or alternatively formed integral with the actuation ring 72 and/or the vane arm 74. In some embodiments the ball member 102 is made integral to the actuation ring 72 and/or the vane arm 74 via one or more of casting, additive manufacturing, sintering, or machining, and may be attached by bolting, brazing, glueing, or welding, for example. The link end 82 includes a complimentary socket 106 to receive the ball member 102 thus securing the ball member 102 to the vane link 78. Similar to the vane link 78 having the spherical bearing configuration a link length 92 may be adjustable at one or both of the link ends 82.

[0043] Use of the spherical bearing member 88 or the ball joint in the actuation assembly 70 connecting the vane arm 74, the vane link 78 and the actuation ring 72 provides a snug connection with improved mobility and reduced play or looseness in the joint, so that the risk of fretting is eliminated or greatly reduced. Additionally, the vane link 78 is adjustable to ensure all of the vanes 52 can be individually adjusted to the correct angle, resulting in a more efficient system, instead of manufacturing tolerances dictating the angle of each vane 50.

[0044] The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

[0045] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0046] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A variable pitch stator vane stage of a gas turbine engine, comprising: a plurality of stator vanes extending across a flowpath of the gas turbine engine; and an actuation assembly operably connected to the plurality of stator vanes, including: an actuation ring extending circumferentially around an engine central longitudinal axis of the gas turbine engine; and a plurality of vane links, each vane link operably connecting a corresponding stator vane of the plurality of stator vanes to the actuation ring, a vane link of the plurality of vane links including: a link body; and a first link end connected the link body, the first link end including a first spherical connection to the actuation ring; wherein circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes; and a vane arm operably connected to a vane stem of a vane of the plurality of vanes via a vane bolt and extending from the vane stem, the vane arm secured to a second link end of the vane link, the second link end including a spherical connection to the vane arm.

2. The variable pitch stator vane stage of claim 1, wherein the first link end includes: a first eye; and a first spherical bearing element disposed in the first eye; wherein the vane link is connected to

the actuation ring through the first spherical bearing member.

3. The variable pitch stator vane stage of claim 2, further comprising a first link fastener extending through the actuation ring and the first spherical bearing member thereby connecting the vane link to the actuation ring.

4. The variable pitch stator vane stage of claim 1, wherein a length of the vane link is adjustable.

5. The variable pitch stator vane stage of claim 4, wherein the length of the vane link is adjustable via a threaded connection of the link body to the first link end.

6. (canceled)

7. The variable pitch stator vane stage of claim 1, wherein the second link end includes: a second eye; and a second spherical bearing member disposed in the second eye; wherein the vane link is connected to the vane arm through the second spherical bearing member.

8. The variable pitch stator vane stage of claim 7, further comprising a second link fastener extending through the vane arm and the second spherical bearing member thereby connecting the vane link to the vane arm.

9. A compressor section of a gas turbine engine, comprising: one or more rotor stages configured to rotate about an engine central longitudinal axis; one or more stator stages, the one or more stator stages including a variable pitch stator vane stage including: a plurality of stator vanes extending across a flowpath of the gas turbine engine; and an actuation assembly operably connected to the plurality of start stator vanes, including: an actuation ring extending circumferentially around the engine central longitudinal axis of the gas turbine engine; and a plurality of vane links, each vane link operably connecting a corresponding stator vane of the plurality of stator vanes to the actuation ring, a vane link of the plurality of vane links including: a link body; and a first link end connected the link body, the first link end including a first spherical connection to the actuation ring; wherein circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes; and a vane arm operably connected to a vane stem of a vane of the plurality of vanes via a vane bolt and extending from the vane stem, the vane arm secured to a second link end of the vane link, the second link end including a spherical connection to the vane arm.

10. The compressor section of claim 9, wherein the first link end includes: a first eye; and a first spherical bearing member disposed in the first eye; wherein the vane link is connected to the actuation ring through the first spherical bearing member.

11. The compressor section of claim 10, further comprising a first link fastener extending through the actuation ring and the first spherical bearing member thereby connecting the vane link to the actuation ring.

12. The compressor section of claim 9, wherein a length of the vane link is adjustable.

13. The compressor section of claim 12, wherein the length of the vane link is adjustable via a threaded connection of the link body to the first link end.

14. (canceled)

15. The compressor section of claim 9, wherein the second link end includes: a second eye; and a second spherical bearing member disposed in the second eye; wherein the vane link is connected to the vane arm through the second spherical bearing member.

16. The compressor section of claim 15, further comprising a second link fastener extending through the vane arm and the second spherical bearing member thereby connecting the vane link to the vane arm.

17. A gas turbine engine, comprising: a combustor section; a turbine section driven by products of the combustion section; a compressor section driven by rotation of the turbine section; and a variable pitch stator vane stage disposed at at least one of the turbine section and the compressor section, the variable pitch stator vane stage, including: a plurality of stator vanes extending across a flowpath of the gas turbine engine; and an actuation assembly operably connected to the plurality of stator vanes, including: an actuation ring extending circumferentially around an engine central

longitudinal axis of the gas turbine engine; and a plurality of vane links, each vane link operably connecting a corresponding stator vane of the plurality of stator vanes to the actuation ring, a vane link of the plurality of vane links including: a link body; and a first link end connected the link body, the first link end including a first spherical connection to the actuation ring; wherein circumferential movement of the actuation ring about the engine central longitudinal axis urges rotation of the plurality of vanes about their respective vane axes; and a vane arm operably connected to a vane stem of a vane of the plurality of vanes via a vane bolt and extending from the vane stem, the vane arm secured to a second link end of the vane link, the second link end including a spherical connection to the vane arm.

18. The gas turbine engine of claim 17, wherein the first link end includes: a first eye; and a first spherical bearing member disposed in the first eye; wherein the vane link is connected to the actuation ring through the first spherical bearing member.

19. The gas turbine engine of claim 18, further comprising a first link fastener extending through the actuation ring and the first spherical bearing member thereby connecting the vane link to the actuation ring.

20. The gas turbine engine of claim 17, wherein a length of the vane link is adjustable via a threaded connection of the link body to the first link end.
